

Nurturing the Next-Generation Maritime Workforce

*Changes & Challenges in Preparing for Alternative Fuels,
New Technologies, and SMR Integration*

대체연료 · 신기술 · SMR 도입에 대비한 차세대 해운 인력 양성



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Why the Workforce Question — Now

기술은 도입돼도, 그것을 운용할 사람은 가장 느리게 준비된다



Alternative Fuels

대체연료

Ammonia, methanol, hydrogen — toxic & flammable handling demands new competencies.



New Technologies

신기술

MASS, digital twins & automation shift the operator's role from manual to supervisory.



SMR Propulsion

소형모듈원자로

Nuclear-powered ships introduce an entirely new licensing & radiation-safety regime.



The convergence point: three transitions arrive at once, but training systems adapt last. A workforce gap is not an HR issue — it is a safety risk.

Every Fuel Transition Has Reshaped the Seafarer

연료가 바뀔 때마다 선원의 역량, 정원, 안전 요건도 함께 바뀌었다

1

to ~1910s

Coal — Solid
석탄 (고체)

Stokers & trimmers;
labour-heavy hand firing.

2

1912 →

Oil — Liquid
석유 (액체)

Diesel era opens; smaller
crews, global bunkering.

3

2012 →

LNG — Gas
LNG (기체)

Gas-fuel safety; first IGF-
Code LNG ships.

4

2020 →

Cleaner Oil
저유황유

IMO 2020 0.50% cap;
VLSFO, scrubbers, MGO.

5

2020s →

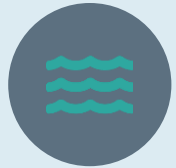
Net-Zero & SMR
탈탄소·원자력

Methanol, ammonia,
hydrogen, nuclear.

Solid → Liquid (1912) → Gas (2012): each leap took ~100 years. The net-zero shift is the steepest skills jump of them all.

From Sail to Coal — The First Industrial Crew

범선에서 석탄 증기로 — 노동집약적 기관 인력의 등장



Age of Sail (to ~1860s)

- Propulsion was wind; skill lived in seamanship, rigging and celestial navigation.
- Early steam engines first served only as auxiliary power on large sailing ships.
- Calm rivers and lakes hosted the earliest steamers before ocean steam matured.



Coal & Steam (1860s–1910s)

- Screw propeller + better marine steam engines made ocean steamships standard.
- Coal demanded large 'black gang' crews — stokers and trimmers shovelling by hand.
- Refuelling needs drove a global network of coaling stations and colonial ports.

Lesson: even the first fuel shift was as much about manning and infrastructure as about the engine.

The Oil Revolution — Speed, Range, Fewer Hands

석유·디젤로의 전환 — 속도·항속거리 향상과 정원 감축

- US oil production began in 1859; through the 1900s ships steadily converted coal → oil.
- 1912 — M/S Selandia (Burmeister & Wain) entered service, the first large ocean-going diesel merchant ship.
- Churchill's Royal Commission on Fuel Oil (1912) led the Royal Navy to its 1913 all-oil policy.
- Queen Elizabeth-class & USS Nevada-class (1912) were the first all-oil-fired capital ships.
- Oil's higher energy density freed cargo space and ended hand-firing — far smaller engine crews.
- But in 1912 the world had barely one liquid-fuel supply port — the fuel led, infrastructure lagged.

1912

M/S Selandia: first large ocean-going diesel merchant ship

-50%

Stokers Fisher expected oil to save vs coal

~1

Liquid-fuel supply ports worldwide in 1912

When Regulation Became the Fuel Driver

환경 규제가 연료를 이끈다 — MARPOL · IMO 2020 · 그리고 Net-Zero 로드맵

1973

MARPOL adopted

IMO's pollution-prevention convention is born.

2012

3.50% sulphur cap

First global HFO sulphur ceiling takes effect.

2018

First LNG-fuelled bulker

Ilshin Green Iris (HMD, IGF Code) — LNG era at sea.

2020

0.50% 'IMO 2020'

Sharp cut from 3.5%; ECAs already at 0.10%.

2023→

CII + EU ETS

Carbon-intensity rating; shipping enters EU ETS.

The road past 2020 — tightening toward net-zero

- EU ETS phases in: 40% (2024) → 70% (2025) → 100% (2026) of shipping emissions priced.
- FuelEU Maritime GHG-intensity limits begin 2025, tightening to -80% by 2050.
- IMO 2023 strategy: net-zero GHG by ~2050, with 2030 / 2040 checkpoints.

Sources: IMO 'Sulphur 2020' & 2023 GHG Strategy; MARPOL Annex VI; EU ETS / FuelEU Maritime; LR & Korean Register (Ilshin Green Iris).

Decarbonisation: Many Fuels, One Workforce Gap

탈탄소 전환 — 연료는 여럿, 그러나 인력 격차는 공통의 과제



LNG / LPG

Cryogenic handling, gas-fuel safety, methane-slip control.



Methanol

Low flashpoint, toxicity; growing dual-fuel newbuild orders.



Ammonia

Highly toxic; demanding training & safety systems.



Hydrogen

Storage & bunkering immaturity; fuel-cell competence.



SMR / Nuclear

Reactor & radiation safety; a wholly new licence class.

Past transitions changed one fuel for another. This one runs several fuels at once — each with new toxicity, safety and licensing demands. The common bottleneck is no longer the engine; it is the trained seafarer.

Why Marine Fuel Keeps Changing

연료 전환의 4대 동인 — 어느 시대든 같은 힘이 작동했다



Energy density 에너지 밀도

Oil held ~2× the heat of coal — smaller bunkers, twice the range, more cargo space. Every later fuel is judged the same way.



Cost & logistics 비용·물류

Fuel must be priced on energy content, not per tonne, and needs a bunkering network. In 1912 there was ~1 oil port; alt-fuels face the same gap.



Environment & regulation 환경·규제

SOx/NOx then GHG rules (MARPOL → IMO 2020 → EU ETS / net-zero) now set the pace more than price does.



Manning & safety 정원·안전

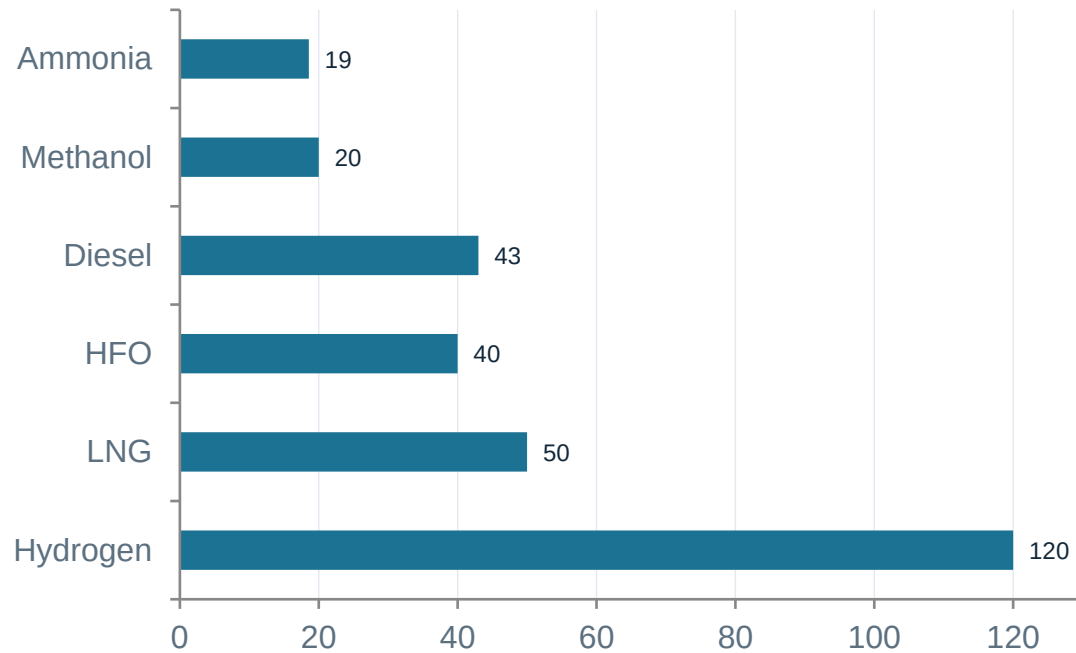
Coal needed armies of stokers; oil cut them sharply. New fuels reverse this — toxicity & cryogenics demand more training, not less.

Historically cost & energy density led; today environment & regulation lead — but a fuel only wins when all four line up.

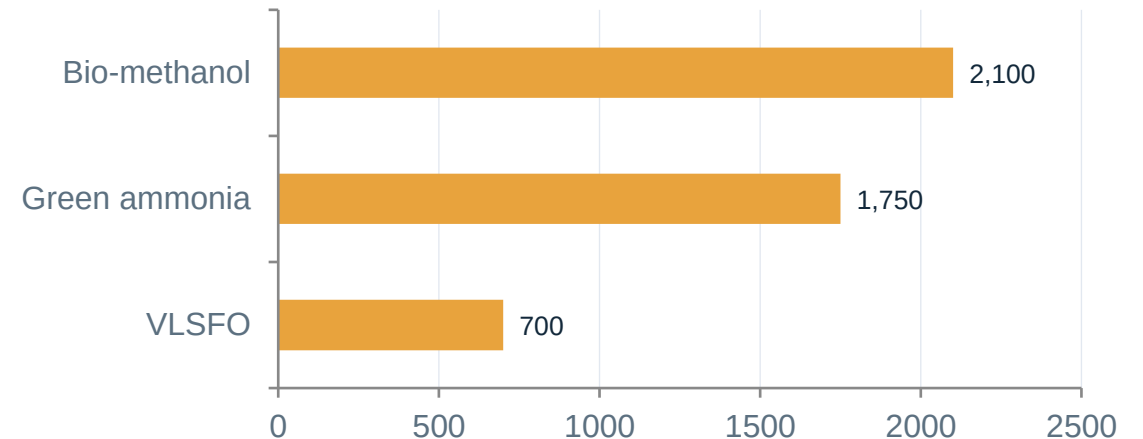
The Drivers in Numbers

객관적 데이터 — 에너지 밀도와 비용이 연료 선택을 좌우한다

Gravimetric energy density (MJ/kg)



Cost, VLSFO-equivalent (USD/ton, 2023–24)



- Energy-adjusted: ammonia holds ~33% and methanol ~36% of LNG's energy per tonne.
- Closing the price gap to VLSFO implies a carbon price of roughly \$300–350+/tCO₂.
- EU ETS adds ~€320/ton to VLSFO at €100/tCO₂ — regulation, not spot price, drives the switch.

SMR at Sea Is Not New — History's Lesson

선박 원자로는 이미 검증됐다 — 실패한 곳은 원자로가 아니라 사람·조직·제도였다

Four civilian reactor ships built (1960s–80s)

- 1962** **NS Savannah (USA)**
454,675 mi on 163 lb uranium; technical success.
- 1968** **NS Otto Hahn (W. Germany)**
126 voyages, ~650,000 nm with no technical fault.
- 1969** **NS Mutsu (Japan)**
Radiation-leak incident; never carried cargo.
- 1988** **Sevmorput (USSR/Russia)**
Only nuclear cargo ship still operating today.



Case — NS Mutsu (1974)

Reactor shutdown safely; failure was organisational, not nuclear

- Almost no experienced reactor-shield designers in Japan at the time
- Westinghouse warned of neutron 'streaming' — warning was not acted on
- Hull & reactor ordered to separate firms; shielding never integrated
- Frequent reorganisation; unclear responsibility & incomplete design review

Reactors proved reliable at sea (Otto Hahn: 650,000 nm, no fault). Where nuclear shipping failed — Mutsu's shielding, and the port bans on Savannah & Otto Hahn — the cause was a competency, organisational and regulatory gap. That is exactly the workforce risk an SMR-ready Korea must close first.

SMR or MSR? Clearing Up the Nuclear Terms

SMR은 '크기·배치' 분류, MSR은 '노형' 분류 — 둘은 경쟁 개념이 아니다



SMR

Small Modular Reactor · 소형 모듈 원자로

A size & deployment class: compact, factory-built, scalable units (typically ≤ 300 MWe). Says how big and how built — not which physics.



MSR

Molten Salt Reactor · 용융염 원자로

A reactor type (Gen-IV): molten salt as coolant/fuel, near-atmospheric pressure, strong passive safety. Says how the reactor works.

An SMR can be built as any of these reactor types:

PWR

Pressurised water — proven, naval heritage

MSR

Molten salt — low pressure, marine favourite

SFR / LFR

Sodium / lead fast — high efficiency

HTGR

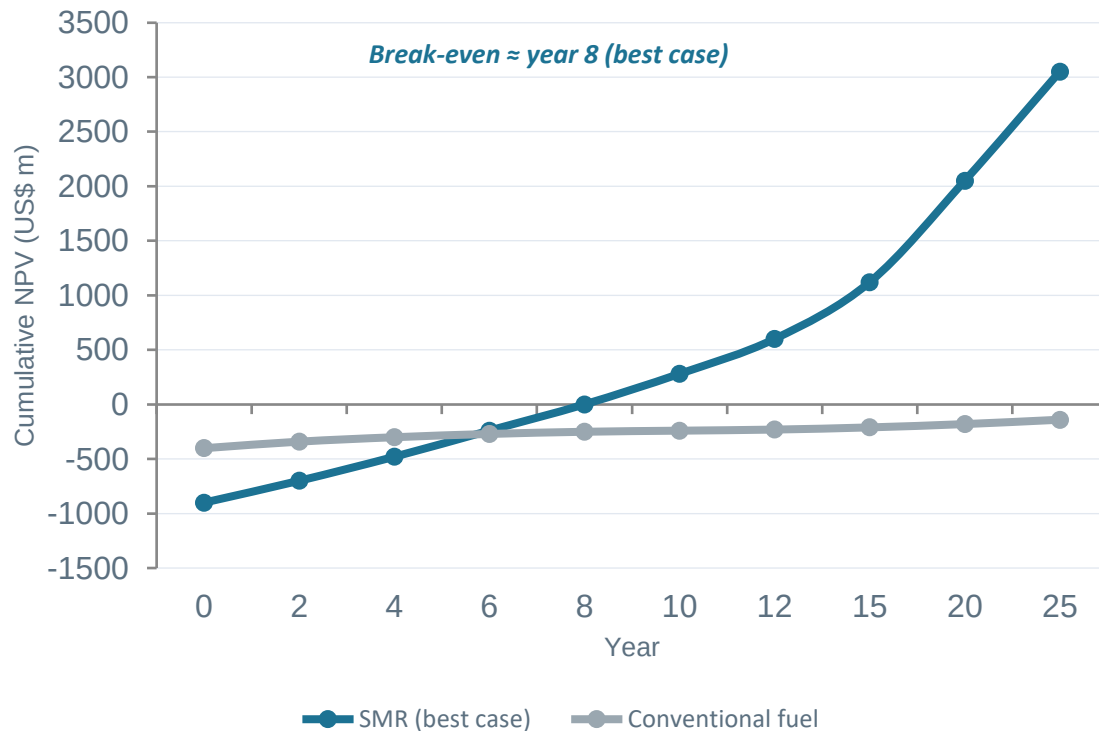
Gas-cooled — high temperature

For this talk: say "SMR" as the umbrella term, and name "MSR" as the leading reactor type for ships — **Korea (KAIST/UNIST, Samsung Heavy, HD Hyundai) and CorePower / China's TMSR all back molten salt for marine.**

Does the Money Work? SMR Break-Even

SMR 추진선의 손익분기 — 높은 초기 CAPEX, 낮은 운항비로 회수

Cumulative NPV — 15,000 TEU containership (illustrative)



~8 yrs

Break-even / payback, best case (Kim et al. 2025, 15,000 TEU, ~70 MWe SMR)

\$68 M

Annual OPEX eliminated: ~\$50M fuel + ~\$18M carbon penalties (LR / LucidCatalyst, Seaspan, 2025)

\$750–
1,000

Possible shipborne SMR cost per kW — well below conventional nuclear (NucNet, 2025)

+38% /
+39%

Higher annual cargo throughput / speed at 25 kn vs conventional

Decisive levers: reactor CAPEX, speed, and carbon price — nuclear gets cheaper per voyage the faster it runs, the opposite of oil.

Sources: Kim et al., Int. J. Naval Arch. & Ocean Eng. 2025 (break-even ~yr 8); Lloyd's Register / LucidCatalyst for Seaspan 2025 (\$68M/yr); NucNet 2025; ABB / OSM 2025 (payback 10–15 yrs).

How an SMR Ship Contains Radiation

심층방어(Defence-in-Depth) — 누출은 '막는' 게 아니라 '여러 겹으로 가둔다'

Three physical barriers (PWR/SMR)

1

Fuel cladding Seals fission products inside each fuel rod.

2

Primary coolant circuit Sealed reactor pressure boundary.

3

Containment structure Withstands a coolant-pipe rupture.

+

Ship reactor compartment Double hull + biological shielding around it.

Tiered safety + marine-specific measures

- Tier 1 — Safety by design: low power & pressure (esp. MSR) can practically eliminate large-release scenarios.
- Tier 2 — Passive systems: natural circulation, gravity feed; cool down with no operator action or power.
- Tier 3 — Active systems + emergency response: shutdown, isolation, radionuclide suppression.
- Marine-specific: collision/fire/flooding cases, ship-motion margins, secure shutdown if the hull is lost.
- Operational: exclusion/controlled zones onboard, dosimetry, trained radiation-protection crew, port emergency plans.

The real gap is not physics — it is rules. No adequate international regime yet covers merchant-ship reactors: liability (land-based conventions don't fit), port access, and transit oversight all need new IMO/IAEA frameworks.

SMR Merchant Ships — Who Regulates What

국가·국제 기구별 SMR 상선 법령 비교 — 아직 어느 곳도 완성된 체계는 없다

Jurisdiction	Key instrument / status	Maturity
IMO (international)	SOLAS Ch. VIII + 1981 Nuclear Code (A.491) — PWR-only; draft new code approved Nov 2025, ~2-yr feedback.	Revising
IAEA (international)	ATLAS project (licensing for sea, results ~2027–28); NEMO became IMO NGO 2025 to harmonise standards.	Developing
United Kingdom	Merchant Shipping (Nuclear Ships) Regulations 2022 — first explicit modern national rules; LR/Core Power policy framework (2025).	Enacted
South Korea	SMR Special Act 2026 (land SMR); K-Moonshot: MSR basic design 2029 → build 2035. Marine reactor rules still a gap.	Building
United States	NRC licenses the reactor; no merchant-ship-specific code. Policy push (2025) + ABS AiP for nuclear ships.	Early
China / Japan / Norway	China: thorium-MSR & floating-NPP concepts. Japan: DNV-classed concept designs. Norway: NuProShip Gen-IV study.	Concept

Common pattern: national reactor licensing exists, but no jurisdiction yet pairs it with a merchant-ship safety code, port-entry rules and cross-border liability. The flag state grants the licence; bilateral deals enable port calls — until IMO's revised Nuclear Code lands (~2027–28).

Sources: World Nuclear Association (2025); Core Power; Nuclear Eng. Int'l (IMO/IAEA/ATLAS/NEMO); J. Marine Sci. & Eng. (UK 2022 regs); NEI / UPI (Korea SMR Act, K-Moonshot); gCaptain; ABB (2025).

The Reality Today — What Actually Sails

전 세계 현행 원자력 추진 선박 실태 — 상업 상선은 사실상 0척, 운항선은 러시아 쇄빙선단

0

Commercial nuclear merchant ships in active service (Sevmorput laid up / retiring)

8

Russian nuclear icebreakers in service — the only operating civil fleet

4

Nuclear merchant ships ever built: Savannah, Otto Hahn, Mutsu, Sevmorput

~400

Reactor-years of Russian Arctic civil-ship operating experience (to 2021)

Civil merchant ships (all 4 ever built)

- NS Savannah (USA, 1962) — museum ship since 1972; PWR.
- NS Otto Hahn (W. Germany, 1968) — re-engined to diesel 1979.
- NS Mutsu (Japan, 1969) — 1974 shielding leak; retired 1992.
- NS Sevmorput (USSR/Russia, 1988) — last in service; laid up, replacement by diesel planned.

Active fleet — Russian nuclear icebreakers

- 8 in service (Rosatom/Atomflot): Arktika, Sibir, Ural, Yakutia, Yamal, 50 Let Pobedy, Taymyr, Vaygach.
- Project 22220 (RITM-200 PWR, 175 MWt) leads the new generation; 40-yr service life.
- 3 more building → ~11 by 2035; larger Rossiya (RITM-400) due 2027.
- Used only in the Arctic, escorting merchant convoys on the Northern Sea Route.

The lesson for the SMR era: civil nuclear propulsion has worked safely at scale — but only in one niche (Arctic icebreakers) under one state operator. No open-market merchant fleet exists yet. The technology is proven; the commercial, regulatory and crewing ecosystem is not.

Is It Legal on International Voyages?

국제항해 원자력 상선 — 국제법상 '항행은 가능, 그러나' 상업 운용은 미완

✓ WHY IT IS POSSIBLE · 가능한 이유

- UNCLOS Art. 17: ships of ALL states — incl. nuclear-powered — have the right of innocent passage.
- UNCLOS Art. 23: foreign nuclear ships may pass if they carry documents & observe special precautions set by international agreements (it expressly contemplates them).
- UNCLOS Arts. 21(2)/22: coastal states can route them to sea lanes but can't bar them by ship type alone.
- Flag-state primary jurisdiction + SOLAS Ch. VIII & the 1981 Nuclear Code give an actual safety framework.
- Proven in practice: NS Savannah & Otto Hahn made international voyages via bilateral agreements.

✗ WHY IT IS NOT YET PRACTICABLE · 가능하지 않은 이유

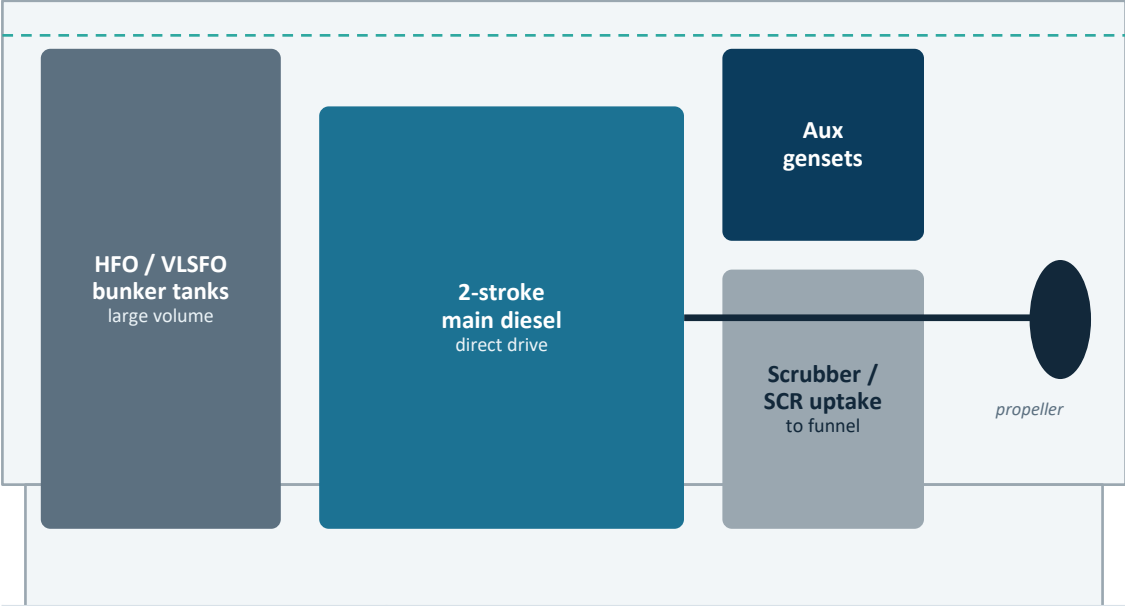
- No liability regime in force: the 1962 Brussels Nuclear Ships Convention never entered into force; Vienna 1963 excludes ship reactors.
- The 1981 Nuclear Code covers PWRs only — not SMR / MSR / floating plants.
- The 1968 IMO/IAEA ports guidance was never revised → effectively invalid.
- Innocent passage ≠ port entry: a coastal state may refuse port access and may suspend passage for security.
- No harmonised port-access or transit-oversight rules → reliance on ad-hoc bilateral deals; ratification deadlock persists.

Bottom line: international law already permits a nuclear merchant ship to sail — the blocker is the missing liability + port-access + transit framework. Closing that gap is squarely IMO/IAEA work, where Korea's delegation can lead.

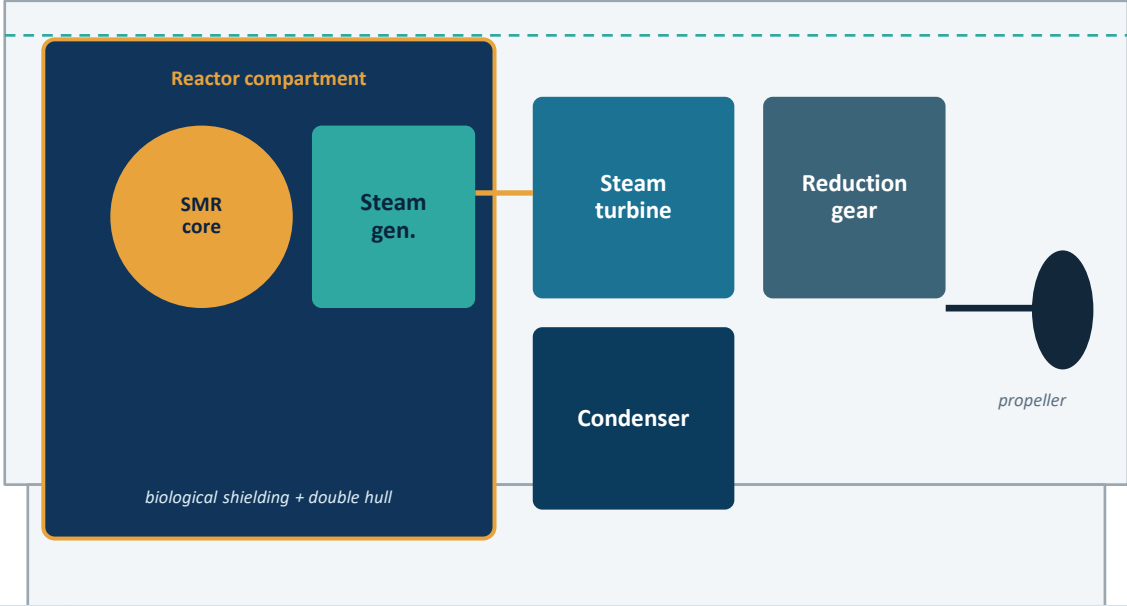
Engine Room: Diesel vs Nuclear (SMR)

기관실 배치·기기 비교 — 개념 단면도 (schematic, not to scale)

Conventional diesel motorship



Nuclear (SMR) ship

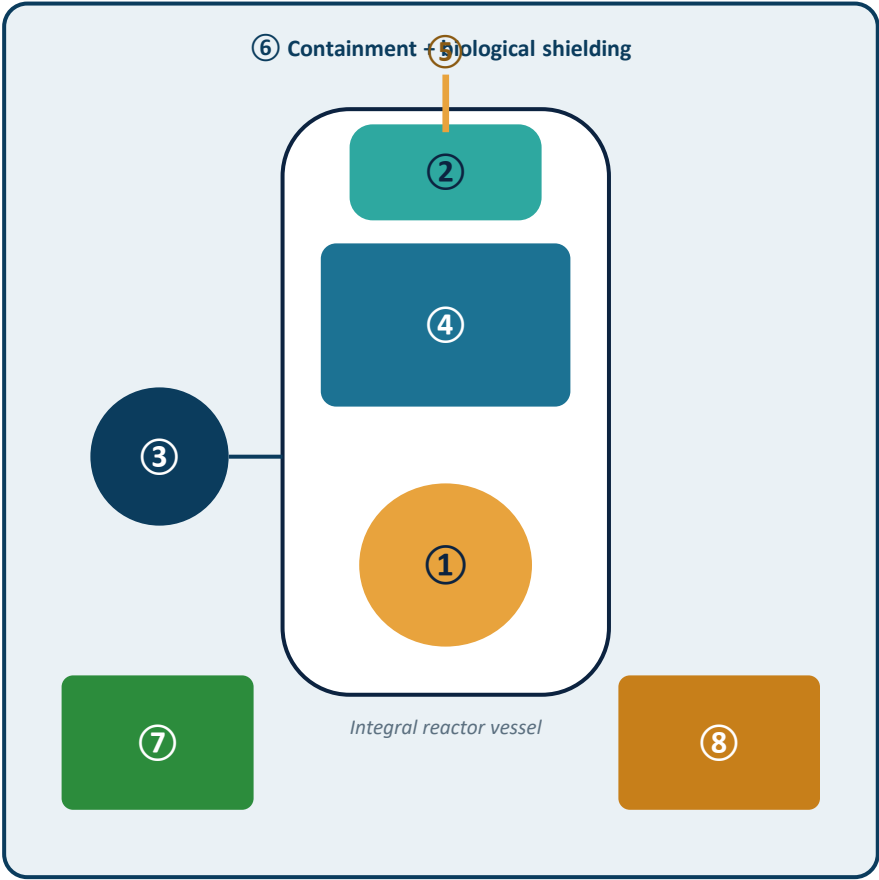


	Diesel motorship	Nuclear (SMR) ship
Prime mover	2-stroke diesel, direct to shaft	Reactor → steam → turbine → gear/shaft
Fuel / energy store	Large HFO/VLSFO bunker tanks; frequent bunkering	Compact core; no refuel for ~25 yrs (MSR) — tanks freed
Signature kit	Scrubber/SCR, exhaust uptakes, funnel	Shielded containment, steam generator, condenser; no funnel
Crew & safety	Engineers; fire/oil-spill regime	+ radiation-protection officers, dosimetry, exclusion zone
Space & weight / MW	Baseline (engine + big bunkers)	Reactor block similar—larger (shield), but no bunkers/funnel → propulsion+fuel TOTAL smaller & lighter, +~5% cargo

Schematic only — not to scale, not a class drawing. U-235 ≈ 8×10⁷ MJ/kg vs HFO ~40 MJ/kg, so the core is tiny; shielding adds weight back but deleting bunkers/uptakes/scrubber makes the total more compact. Sources: US Navy/FAS, SubGuru, NavWeaps, ANSTO, Nautilus Shipping, LR/Seaspan.

Must-Have Equipment on a Nuclear Ship

원자력 추진 상선 필수 기기와 역할 — 일체형 PWR/SMR 개념도 (schematic)



Schematic only — not to scale, not a class drawing.

- ① **Reactor core & fuel** — Controlled fission generates heat; the energy source.
- ② **Pressuriser** — Holds primary water at ~150 bar so it can't boil.
- ③ **Reactor coolant pumps** — Circulate primary coolant core → steam generator.
- ④ **Steam generator** — Heat-exchanger: primary heat → clean secondary steam.
- ⑤ **Control rods + drives** — Absorb neutrons; raise/lower power & SCRAM shutdown.
- ⑥ **Containment + shielding** — Seals fission products; biological shield cuts dose.
- ⑦ **Emergency core cooling / decay-heat removal** — Passive + active cooling after shutdown (LOCA safety).
- ⑧ **Radiation monitoring & dosimetry** — Continuous dose/contamination watch across the ship.

Initial IMO framework (1981 Code) covers PWRs; SMR/MSR designs follow the same safety functions. Sources: SOLAS Ch. VIII / 1981 Code of Safety for Nuclear Merchant Ships (Res. A.491); MDPI J. Nucl. Eng. 'Review of Maritime Nuclear Reactor Systems' 2025; MIT Maritime Consortium Nuclear Ship Safety Handbook 2025; World Nuclear Association.

SOLAS Ch. VIII — How Far It Lets You Sail

SOLAS 협약상 원자력 추진 상선의 국제항해 허용 범위와 조건

Starting point: SOLAS applies to ships on international voyages — and Ch. VIII applies to ALL nuclear ships except warships. So international trade by a nuclear merchant ship is contemplated, not prohibited.

What Ch. VIII requires (the green light & its conditions)

Reg. 1–2	Applies to all nuclear ships (not warships); other SOLAS chapters still apply as modified.
Reg. 6	Administration must ensure no unreasonable radiation hazard — at sea or in port.
Reg. 7	Safety Assessment prepared & supplied in advance to countries the ship intends to visit.
Reg. 9–10	Special survey at least yearly → Nuclear Cargo/Passenger Ship Safety Certificate (max 12 months).
+ Code	1981 Code of Safety for Nuclear Merchant Ships (Res. A.491) supplements all the detail.

Where the permission stops (the limits)

Reg. 11	Nuclear ships face SPECIAL CONTROL before & in port — a port may verify the certificate and refuse entry if hazards aren't acceptable.
PWR-only	The 1981 Code covers pressurised-water reactors only — SMR/MSR/FNPP not yet inside its scope.
Port ≠ passage	SOLAS sets ship-safety, not a right of port access; entry still rests on the visited state's acceptance of the Safety Assessment.
No liability	SOLAS does not provide a nuclear-damage liability regime — that gap (Brussels 1962, unratified) remains separate.

Scope in one line: SOLAS Ch. VIII permits a (PWR) nuclear merchant ship to trade internationally if it holds a valid certificate and a pre-shared Safety Assessment — but each port state keeps a special-control veto, and SMR/MSR await the revised Code (approved in draft, Nov 2025).

Port Entry for a Nuclear Ship — Russia's Practice

러시아 원자력 추진 선박 입항 — 필요 문서와 절차 (국제 틀 + 러시아 자국 규제)

Documents the ship must carry / pre-share

- 1 Valid Nuclear Cargo Ship Safety Certificate (SOLAS Reg. 10; max 12 months).
- 2 Safety Assessment — prepared & supplied IN ADVANCE to the visited state (SOLAS Reg. 7).
- 3 Reactor & operating documentation: operating manual, radiation-survey records, dosimetry log.
- 4 Flag-state / national-regulator authorisation (in Russia: Rosatom / Atomflot operator approval).
- 5 Advance notification + last-10-ports-of-call list; FSB security clearance (2025 Russian rule).

Procedure — from request to acceptance

- 1 **Advance notification** — Operator/flag state notifies the port state ahead of arrival, with the Safety Assessment.
- 2 **Safety-assessment review** — Receiving state evaluates the reactor & radiation case (SOLAS Reg. 7) and decides acceptance.
- 3 **Special control (Reg. 11)** — Before & in port: verify valid certificate + confirm no unreasonable radiation hazard.
- 4 **Security / underwater check** — Russia 2025: FSB clears within 48 h of the 1-hour port-captain notice; diver inspection possible.
- 5 **Accept or refuse** — Port grants entry — or refuses (as Far-East ports did to Sevmorput in 1988).

Reality: Russia runs its own Arctic/NSR ports under one operator (Atomflot/Rosatom), so domestic calls are routine — but international calls still hinge on the visited state's acceptance + a special-control veto. Sevmorput was refused by its own Far-East ports in 1988, and a 1990 request to call Vancouver was denied. There is no harmonised international port-entry regime — only bilateral acceptance.

Where Nuclear Propulsion Actually Fits

원자력 추진 상선의 현실적 적합 영역 — 톤수·선종·항해유형



Power / size 출력·규모

Reactor power must match the ship. Large ocean ships need >50 MWe; small ships <5 MWe. Real concepts: HD Hyundai 16,000 TEU (~100 MW), Suezmax tanker 155,000 DWT (70 MWt). Fixed reactor cost favours BIG, high-power hulls.



Ship type 선종

Best fit: large container ships, crude/product tankers, bulk & ore carriers, LNG carriers, ice-class Arctic ships, cruise. These run high, steady power on long legs — where nuclear's no-refuel endurance pays.



Voyage profile 항해유형

Deep-sea / ocean-going, fixed long-haul routes (e.g. Asia–US, Asia–Europe). NOT short-sea/coastal feeders: frequent port calls multiply the special-control & port-acceptance friction and waste the long-endurance core.

✓ Well suited

Large ($\approx$ >50,000 DWT or >10,000 TEU), high-power, fast deep-sea liners and bulk/tanker trades on fixed intercontinental routes; ice-class Arctic ships. Long legs + few calls = nuclear's endurance and freed cargo space win.

✗ Poorly suited (for now)

Small coastal & short-sea feeders, ferries, frequent multi-port callers: low power needs, repeated port-entry/special-control friction, and high fixed reactor cost per ship make the economics and access hard.

Design intent: the realistic first SMR merchant ships are large, high-power, ocean-going liners/tankers/bulkers on long fixed routes — not coastal feeders. Korea's 15,000–16,000 TEU SMR container concept sits squarely in that sweet spot.

Built — But Still Refused Entry? Why

원자력 추진 상선을 건조해도 입항 거부될 수 있는 이유

1

Port-state sovereignty — no right of port entry 항만국 주권 · 입항권 없음

A port state may set entry requirements above international standards, designate which ports are open, and close them on national-interest grounds. Innocent passage ≠ a right to enter port (UNCLOS Arts. 25(2)/211(3); The National Interest 2025).

2

No liability / insurance regime in force 책임·보험 체계 부재

The 1962 Brussels Nuclear Ships Convention never entered force; P&I clubs & nuclear-risk insurers want a multilateral treaty before underwriting. Without clear cross-border liability, a port can refuse to accept the exposure.

3

Emergency Planning Zone (EPZ) clash 비상계획구역(EPZ) 충돌

Land-reactor EPZ criteria applied to a moving reactor impose unacceptable constraints at busy commercial ports — and may require evacuation/response capability the port cannot show (cf. 1990 Vancouver denial of Sevmorput).

4

Public opposition & political risk 대중 반대·정치적 위험

Local protest and radiation fear can block a call regardless of paperwork — Sevmorput was refused by its own Far-East ports in 1988 and harbour workers refused to handle it (post-Chernobyl).

5

Certificate / reactor-type mismatch 증서·노형 불일치

Special control (SOLAS Reg. 11) can refuse a ship lacking a valid certificate or an accepted Safety Assessment; the 1981 Code is PWR-only, so an SMR/MSR may fall outside what a port will currently recognise.

Key point: building the ship is necessary but not sufficient. Market access depends on liability/insurance, EPZ rules, public acceptance and a recognised certificate — all decided by each port state. This is the case for harmonised IMO/IAEA rules, where Korea can lead.

Clearing the Path — Institutions & People

해결책 통합 — 제도를 고치고(좌) 사람을 길러(우) 원자력 전환을 실현한다

Fix the institutions · 제도

PWR-only Code

Finalise the technology-neutral revised IMO Nuclear Code (IMO + IAEA ATLAS).

No liability regime

New multilateral nuclear-ship liability treaty + P&I / nuclear-risk pool.

Port-entry refused

Harmonised acceptance + ship-specific EPZ + green-corridor pilots (bilateral first).

Public trust

Transparent safety case, passive-safe design, independent oversight.

High CAPEX

Large deep-sea hulls; reactor leasing; carbon pricing closes the gap.

Build the workforce · 인력 양성

01

Recognise simulation as sea service

Digital-twin & simulator training under a STCW V/4-style rule (ASME V&V 40; ICAO/EASA precedent).

02

Modular fuel/tech endorsements

Stackable competence modules on the core CoC — no slow wholesale rewrite.

03

Maritime–nuclear curriculum

First seafarer SMR qualification: reactor + radiation safety + bridge/engine, with the regulator.

04

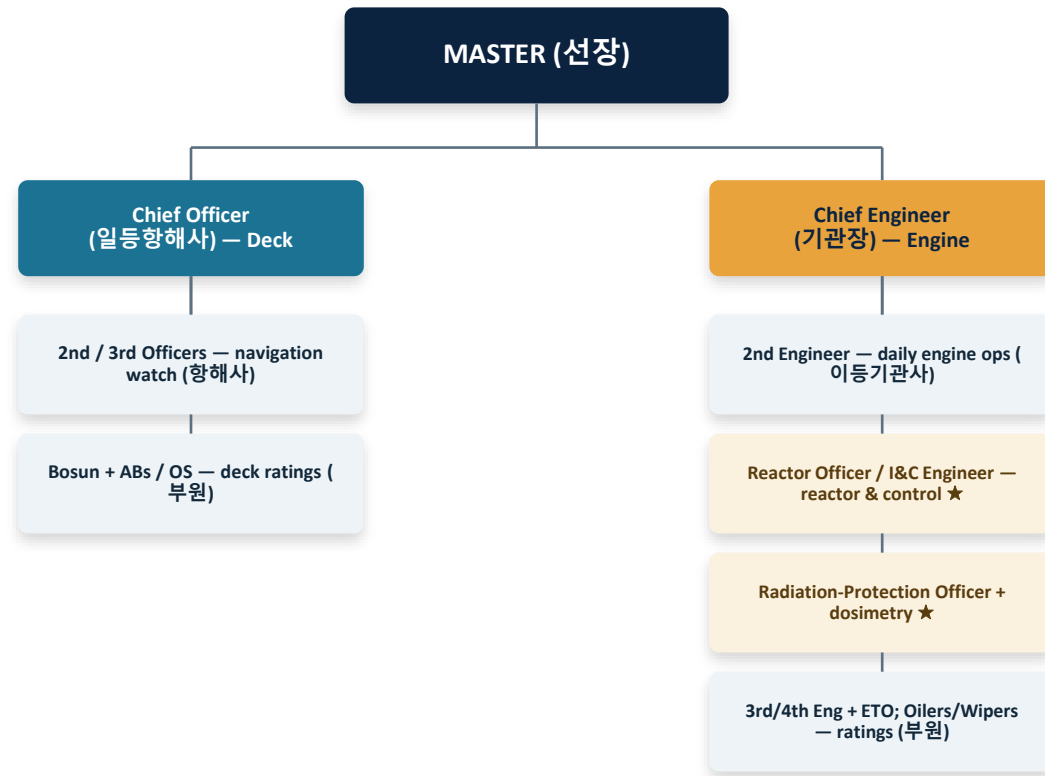
Industry–academia–IMO co-design

Tripartite delivery so standards, simulators & trainers scale together.

The unifying thread: every barrier is institutional, not physics — and people sit at the centre. Korea uniquely builds the ships AND shapes the STCW/IMO rules; leading the maritime-nuclear competency standard turns a follower market into a leadership one.

Crewing the SMR Ship — Proposed Structure

원자력(SMR) 추진 상선 직제 — 직급 상하관계·직무 (제안 모델+ 러시아 실증)



★ = new nuclear roles (no diesel-ship equivalent)

Roles & duties

Master

Overall command, safety, nuclear-security accountability; port/flag/regulator liaison.

Chief Engineer

Heads engine + reactor plant; ultimate technical authority over the propulsion system.

Reactor Officer / I&C Eng. ★

Runs the reactor from the control room: reactivity, coolant, steam plant; SCRAM & emergency.

Radiation-Protection Officer ★

Dose & contamination monitoring, shielding integrity, waste handling, exposure limits.

Deck officers + ratings (부원)

2/O–3/O navigation & watch; Bosun/ABs on deck; oilers/wipers + reactor assistants under 2nd Eng.

Proven model — Russia's Atomflot icebreakers

The only operating civil fleet already runs this structure: chief (mechanical) engineer over the reactor plant, reactor-control / I&C engineers (a rank confirmed in service), and a dosimetry service. Project 22220 automation cut crew to ~75 (vs 189 on older Arktika-class); officers train at Makarov Academy + Atomflot's full-scale reactor simulator.

Design principle: keep the proven Master → Deck/Engine chain of command, and insert a reactor-operations + radiation-protection cadre inside the engine department under the Chief Engineer. New STCW endorsements define their competence — the workforce gap this talk targets.

What STCW Must Amend for Nuclear Ships

STCW 개정 필요 부분 — 원자력(SMR) 추진 상선 해기 역량 체계

Where to amend (STCW architecture)

New Reg. V/5 (proposed)

Add a dedicated 'special training for personnel on nuclear-powered ships' regulation in Chapter V — the same mechanism used for IGF-Code (V/3) and polar (V/4) ships.

Code Part A (mandatory)

New competence tables: mandatory minimum knowledge, understanding & proficiency for nuclear-ship officers + ratings, at management & operational levels.

Code Part B (guidance)

Recommended training programmes, simulator syllabi and refresher intervals to help administrations implement Part A consistently.

Reg. I/2 & I/11 (certs)

New certificate/endorsement type + revalidation rules — a recognised maritime-nuclear endorsement layered on the existing CoC.

Ch. II / III interface

Define how reactor & radiation duties sit alongside existing deck (II) and engineer (III) competences and watchkeeping (Ch. VIII).

What new competences it must define

- Reactor & nuclear-propulsion fundamentals — physics, reactor types (PWR/SMR/MSR), steam-plant operation.
- Radiation protection & dosimetry — dose limits, ALARA, contamination control, shielding, waste handling.
- Reactor operation & control-room watchkeeping — reactivity, coolant, normal + abnormal operations.
- Emergency response — SCRAM, loss-of-coolant, decay-heat removal, on-board radiological emergency drills.
- Nuclear security & safeguards — fuel security, access control, vetting (esp. if higher-enriched fuel).
- Simulator-based qualification — full-scale reactor-control simulator training recognised as competence (and, per your research, as sea service under V/4-style rules).

The amendment path: a new Chapter V regulation + Code A/B tables + an endorsement — proposed and refined through IMO HTW, in step with the revised Nuclear Code. Korea's HTW delegation can table the first draft and lead the competency standard.

What Should a Nuclear-Ship Officer Earn?

러시아 디젤 vs 원자력 해기사 연봉 + HMM 실측·GDP 보정 한국 추정

Russia — diesel vs nuclear officer (master, illustrative)

	RUB / yr	≈ USD / yr
Diesel merchant master (avg)	1.78 M	\$22,200
Nuclear (icebreaker) +40%	2.49 M	\$31,100
Nuclear midpoint +55%	2.76 M	\$34,500
Nuclear (icebreaker) +70%	3.02 M	\$37,800

Premium = specialised reactor cadre + radiation hazard + longer training. Band is an estimate (Atomflot pay is not public).

GDP per-capita factor

Korea \$36,100 ÷ Russia \$14,900 ≈ **2.4×**

Korea — estimate anchored on HMM actual pay

HMM actual (FSS 2023 report): sea-staff avg **₩138 M** (male); company avg ₩123 M — far above the generic ₩79.5 M captain figure (regular-staff, tax-exempt & deferred pay).

Path A — scale Russian nuclear pay by GDP (floor)

$\$34,500 \times 2.4 \approx \$83,500 \rightarrow$ about ₩114 M / yr

Path B — HMM sea-staff base + nuclear premium

₩138 M × +30% → ₩179 M +45% → ₩200 M +60% → ₩221 M

Likely range: ₩140–200 M / yr — GDP-floor to HMM-premium, for a senior nuclear-ship officer.

Reading: anchored on HMM's disclosed sea-staff pay (₩138 M), a senior Korean nuclear-ship officer would plausibly earn ₩140–200 M/yr — the GDP-scaled Russian figure (~₩114 M) is a floor, HMM-base-plus-premium the upper end. Planning estimates, not survey data.

Sources: HMM FSS 2023 annual report (sea-staff avg ₩138 M male / company avg ₩123 M, via News2day); ERI SalaryExpert (Russia captain ₱1.78 M; generic Korea captain ₩79.5 M); IMF per-capita GDP (Korea ~\$36k, Russia ~\$15k). Nuclear premium & FX are assumptions — planning figures only, not Atomflot/employer data.

For the Shipowner — Profit or Loss?

선사 입장의 손익 — 단기 비용 vs 장기 수익, 그리고 손익분기 조건

✗ Costs / risks (short-term) · 불이익

Huge upfront CAPEX — Propulsion cost rises ~10–30× vs conventional; reactor + auxiliary systems dominate the build budget.

Higher crew cost — Specialised nuclear officers earn a +40–60% premium; new reactor/radiation-protection posts added.

Port-access risk — A built ship may still be refused entry (liability, EPZ, public acceptance) — stranded-asset risk.

Regulatory uncertainty — PWR-only Code, no liability regime yet, insurance still maturing — financing risk premium.

✓ Gains / revenue (long-term) · 이익

Fuel + carbon eliminated — Up to ~\$50 M/yr bunker + ~\$18 M/yr carbon removed ≈ \$68 M/yr (15,000 TEU, LR/LucidCatalyst–Seaspan).

More cargo throughput — +38% annual capacity: 25 kn (39% faster, 6.3 vs 5 round voyages) + 5% space from no fuel tanks.

Asset lasts 2× — SMR core life ~50 yrs vs 25-yr hull — one reactor serves two hulls; higher time-utilisation.

Price stability — Fixed long-life fuel insulates from volatile bunker & carbon markets.

Verdict: conditional profit

Short-term it's a cost (CAPEX, crew, regulatory risk); long-term it can be a strong profit — but only when the break-even conditions hold: **high carbon price, large fast deep-sea hulls, reactor leasing to shift CAPEX→OPEX, and resolved port-access/liability rules**. Freight rates drive total cost of ownership; reactor CAPEX matters less. The +40–60% crew premium is small against the ~\$68 M/yr OPEX saved. Payback ~10–15 yrs (OSM).

How Competency Demand Is Changing

세 축이 요구하는 새로운 역량 — 기존 STCW 체계와의 격차(Technology-Regulation Gap)



Alternative Fuels

Fuel-specific toxicity & explosion management · bunkering safety · emissions monitoring (NH₃, CH₃OH, H₂).



Automation & MASS

Remote supervision, sensor-trust judgement, human-autonomy teaming, cyber-resilient operation.



Digital Twin / Data

Model-based operation, condition monitoring, data literacy and verification of simulated systems.



SMR Integration

Nuclear fundamentals, radiation protection, reactor-safety culture & a new statutory licence class.

STCW Chapter II/V competency standards were not written for any of these — the gap widens fastest at the SMR row.

What Stands in the Way

무엇이 인력 양성을 가로막는가



Regulatory lag

STCW revision cycles run years behind technology adoption — endorsements arrive after ships do.



Trainer & infrastructure gap

Too few qualified instructors and simulators for alternative-fuel and autonomous systems.



No SMR qualification path

Maritime nuclear propulsion has no recognised seafarer licence framework anywhere yet.



Unvalidated digital training

Simulator & digital-twin learning lacks an accepted method for recognising it as sea service.

The Takeaway

Whether the green & autonomous transition succeeds will be decided by people — not hulls.

친환경·자율운항 전환의 성패는 결국 사람이 가른다

Recognise digital training

Stack modular competence

Lead maritime-nuclear skills

Thank you · 감사합니다

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